Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	**Developer(s)**	**Change**
October 13, 2026	Andy Huynh	Critical Assumptions
October 15, 2026	Owen Johnson	Introduction, Scope and Purpose Updated
October 17, 2026	Humna Saeed	Section 3: System Boundaries & Components Draft
October 17, 2026	Andreas Sideris	AI Disclosure Updated
October 17, 2026	Christopher Ludwig	Safety and Security Requirements First Draft
October 18, 2026	Andreas Sideris	FMEA Table Completed
October 18, 2026	Andreas Sideris	Reflection Added
October 19, 2026	Owen Johnson	Reflection Added
October 19, 2026	Andy Huynh	Critical Assumptions final
October 19, 2026	Christopher Ludwig	Roadmap and Reflections

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1 Introduction

This document is a hazard analysis of the RealmForge system, which is a roguelite video game. A roguelite game is defined by different attempts to clear the game being varied from each other in some way, and being mostly self-contained with a small amount of certain progression carrying over, with the goal that each attempt feels unique. RealmForge intends to provide an engaging game experience with lots of variability while collecting data from users about the effect of this variability to further enhance the game experience through intuitive usability, fun game design/player experience, and replayability.

Using Nancy Leveson's work (referenced in lecture), a hazard is defined as a property/condition in the system that, when combined with an environment condition, has potential to cause harm or damage. For RealmForge's purposes, **harm** can be described as:

- Player dissatisfaction
- Performance issues
- Poor usability testing data

2 Scope and Purpose of Hazard Analysis

The scope of this document is to provide an analysis of the system as it pertains to potential hazards. The analysis will identify hazards present in the components/boundaries of the system, provide mitigation strategies for each hazard, and provide the resulting safety and security requirements.

This analysis is important, as certain losses can be incurred by the hazards if left unchecked, such as:

- Lower player count and interest
- Player frustration
- Unintended behavior, such as bugs, lag, or crashes
- Lack of usability testing data to work from

3 System Boundaries and Components

The following section identifies the system boundaries of RealmForge and the game's major components. These components help identify and evaluate potential hazards that could affect gameplay, player progress, and the overall functionality of the game.

System Boundaries

RealmForge is a locally running game that allows players to configure game properties, explore different areas, participate in combat encounters, and purchase upgrades using resources earned throughout gameplay. The system is responsible for processing player inputs, displaying gameplay information, updating the game based on player actions, and saving player progress locally.

External components include the user's operating system, input devices such as a keyboard, mouse, or supported controller, and output devices such as a monitor and speakers or headphones.

System Components

Property Configuration

Property configuration allows for the selection and application of game property variations prior to starting a game session. These selections vary based on player preference and influence the overall gameplay experience.

World Navigation

This component handles player movement as they navigate the game world, allowing players to explore different locations as they progress through the game.

Combat System

The combat system is responsible for handling combat encounters between the player and opponents within the game. This includes player attack actions, interactions, and the outcomes of combat encounters.

Progression and Rewards

This component is responsible for tracking resources earned throughout gameplay, as well as player progression and rewards received from completing encounters.

Central Hub

The central hub allows players to purchase items and upgrades using resources earned throughout gameplay. Players return to the hub when starting a new game run, allowing them to prepare for their next session.

Progress Management

The progress management system is responsible for saving player progress locally to ensure that player progress is preserved for future sessions.

User Interface

The user interface component is responsible for displaying gameplay information and allowing player interactions. This includes interactions with menus, combat options, world navigation, the central hub, and other game features.

4 Critical Assumptions

- The application can be correctly installed and launched on a machine running a Windows Operating System (Windows 10 or 11). The application can be closed without causing fatal system crashes in the hardware used.

- The application will not maliciously introduce any computer viruses or malware to the user's machine.
- The application will not cause crashes or hardware meltdowns to input periphals (e.g., keyboard and mouse).
- The contents of the software (in the RealmForge game) will not contain any inappropriate language, or extreme mature themes/images.
- The application will not play any loud audio cues or sounds that can cause damage to the user's hearing (at or exceeding 85dB).

5 Failure Mode and Effect Analysis

Table 2: Failure Mode and Effect Analysis (FMEA)

Component	**Failure Mode**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**Safety Requirements**	**Ref.**
Property Configuration	System is unable to recognize the user's connected input device (such as a controller), when selected in the feature variation menu	Users waste their time and can become confused/frustrated when attempting to troubleshoot input device connection	A. System does not provide sufficient guidance to the user to connect their input device B. System does not correctly monitor device interfaces to receive input (Bluetooth, USB)	A. Provide detailed guidance to the user on how to connect each input device; prompt the user to select the device they wish to connect (PlayStation DualSense controller, Xbox Series X controller, etc.) and provide guidance specific to that device B. Monitor both USB and Bluetooth interfaces for connection to each type of supported input device when the user opts to connect a new device	SSR-010 SSR-011	H1-A

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Table 2 – Continued

Component	**Failure Mode**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**Safety Requirements**	**Ref.**
World Navigation	User is unable to understand where their player character can move next in the overall game world	User can become confused/frustrated, and waste gameplay time staying in the main game world instead of progressing forwards	System does not provide adequate feedback to the user regarding where the player character can move next; valid locations are indistinguishable from invalid locations, or a possible path to take to reach the end of the game is unclear	Highlight valid locations that the player character can move to in the game world, and indicate invalid locations. Provide further guidance to the user regarding a possible path forward if an extended period of time is spent attempting to reach invalid locations	SSR-008	H2-A
Combat System	System displays flashing or intense graphics	Users with photosensitive epilepsy can potentially suffer an epileptic seizure	The system displays excessive flashing graphics without warning users with photosensitive epilepsy	Include a detailed flashing lights or epilepsy warning when the system launches, and ensure the game does not display excessive flashing graphics	SSR-006	H3-A

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Table 2 – Continued

Component	**Failure Mode**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**Safety Requirements**	**Ref.**
Progression and Rewards	Rewards provided to the user for succeeding in an enemy encounter are lost when exiting the encounter to the game world	Gameplay progression is interrupted by earned rewards not being given to the user, preventing them from using rewards to improve their in-game capabilities; this can cause the user to become frustrated	Variables storing enemy encounter rewards are erroneously overwritten by another value during transitions between game scenes	Retain and update enemy encounter reward variables (during encounters) in an additional memory location separate from the combat system, to ensure a backup value is available	SSR-007	H4-A

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Table 2 – Continued

Component	**Failure Mode**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**Safety Requirements**	**Ref.**
Central Hub	System does not return game state and display to the Central Hub menu following a failed game run (user is defeated)	User is unable to continue the correct flow of gameplay (returning to the Central Hub menu, starting again), and may need to restart the software to resolve the issue, causing frustration	Software errors/faults prevent the system from correctly transitioning between game scenes (end of game run/user is defeated to Central Hub menu)	Extensively test and verify major game state transitions (user defeat to Central Hub menu, reaching a game world area to an enemy encounter) for correctness. Game state transitions can be verified against a state machine for clarity	SSR-012	H5-A
Progress Management	System application unexpectedly crashes or closes completely due to an error	Most recent user save data is corrupted and unrecoverable; the user can be frustrated due to lost progress	System does not store any backup user save data	Regularly store user save data in multiple locations; revert to an intact recent saved state on system failure	SSR-004 SSR-005	H6-A

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Table 2 – Continued

Component	**Failure Mode**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**Safety Requirements**	**Ref.**
Progress Management	System attempts to initialize the game using corrupted save data	The system is unable to correctly initialize the game and enable gameplay; the user can be confused on how to proceed using their original saved game data, and can become frustrated	System does not take steps to validate user save data	Validate user save data against previous snapshots; if save data is corrupted, revert to an intact recent saved state	SSR-003	H6-B

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Table 2 – Continued

Component	**Failure Mode**	**Effects of Failure**	**Causes of Failure**	**Recommended Action**	**Safety Requirements**	**Ref.**
User Interface	System displays game and interface graphics that are difficult for colourblind users to interpret	Colourblind users are unable to progress efficiently in the game, causing frustration; the game is not accessible to as many users as possible	Displayed symbols in game and interface graphics are difficult to distinguish for colourblind users (colours such as red and green are used as a primary method of differentiation), or colourblind-accessible graphics filters are not provided	Avoid relying only on certain colours to convey interface and game information (red and green), and provide appropriate colourblind-accessible graphics filter settings to the user	SSR-009	H7-A

6 Safety and Security Requirements

SSR-001: Save data shall be associated with specific game profiles saved on the device.

Rationale: Being able to have multiple game profiles improves usability of the system

Fit Criterion: At least 3 game profiles can be stored on one at device simultaneously, and saving/loading data to/from one profile never affects other saved profiles

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-002: The system shall ensure all play tester information and evaluation metrics are securely stored and restricted from unauthorized access.

Rationale: Play testers should feel comfortable to test the game without fear of having their information exposed to third parties.

Fit Criterion: Tester information never reaches any party outside of the team and unnecessary information is not collected

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-003: Save files and player data shall be validated upon launch, detecting and rejecting malformed data.

Rationale: Losing data or having files be corrupted during launch will result in player frustration.

Fit Criterion: System detects corrupted data ¿99% of the time and never loads corrupted data when detected

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-004: Unsaved progress or session data shall be stored locally on the user's device if an unexpected exception or failure occurs.

Rationale: Losing progress in-game after unexpected exceptions occur will result in immense frustration in users.

Fit Criterion: Unexpected failures never cause the user to lose progress that will take longer than 5 minutes to recover

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-005: The application shall handle unexpected runtime exceptions gracefully to prevent permanent data corruption to local save files or destabilization of the host operating system.

Rationale: Failure of the system resulting in destabilization outside of the game environment is potentially dangerous to host system.

Fit Criterion: Unexpected failures of the system never cause damage or destabilization of the host environment

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-006: The application shall display a warning about flashing lights and not have excessive flashing lights during gameplay.

Rationale: Excessive flashing lights without a warning previously displayed is dangerous for users with epilepsy.

Fit Criterion: Flashing lights warning always displays on start-up

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-007: The system shall store backups of combat reward data and verify user receives these rewards.

Rationale: Not gaining rewards when the user should will negatively impact user experience greatly.

Fit Criterion: Users always receive combat rewards when they should

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-008: The system shall clearly distinguish between actions which are available and unavailable to the user during map navigation, combat, and central hub modification.

Rationale: Player can become frustrated while trying to make relevant decisions if their options are not clearly laid out.

Fit Criterion: Users are aware of all options available to them in 95% of scenarios

Traceability: N/A

SSR-009: The system shall avoid using certain colours to convey important information.

Rationale: The system must be accommodating to users with

colour blindness.

Fit Criterion: Colour blind users do not miss out on any information non-colour blind users can see

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-010: The system shall provide guidance how to connect input devices.

Rationale: Difficulty in connecting input devices can waste user's time and frustrate them during the process.

Fit Criterion: Connecting a new input device takes ¡2 minutes at least 95% of the time

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-011: The system shall detect when new input devices are connected.

Rationale: Makes the process of connecting an input device quick and simple for the user.

Fit Criterion: System always detects when a new input device is connected and notifies the user

Traceability: Created from Recommended Actions devised in FMEA table presented previously

SSR-012: The system shall reconfirm it is in the proper game state after major game state transitions occur

Rationale: The user getting locked out of progressing through game states will prevent playing the game

Fit Criterion: Users never get stuck in certain game states with no way of progressing without restarting the system

Traceability: Created from Recommended Actions devised in FMEA table presented previously

7 Roadmap

All of our safety and security requirements should be able to be implemented as part of this capstone project's timeline without need to postpone any of the requirements until after this project concludes. These requirements all either specify critical data integrity requirements, usability requirements, and data privacy requirements which are absolutely necessary for the ethics of play testing and usability of our game while still remaining simple enough to be implemented within our timeline.

8 Appendix — Generative AI Use Disclosure

AI Tools Used

- OpenAI GPT-6
- Google Search AI Overview (Google Gemini)

How and Where Used

AI tools were used for assistance with LaTeX syntax and commands, during the development of the FMEA table section, and when developing LaTeX formatting for the document as a whole. Specifically, AI tools were used for assistance in creating the LaTeX table used for the FMEA summary and creating the list of tables, as well as resolving various minor formatting bugs. AI tools were also used to create the LaTeX custom requirement format macro used in the document, that was also used in the SRS. In this section (FMEA), AI tools were not used to generate any text or wholly generate any of the ideas or conclusions presented.

Verification of AI-Generated Content

Any LaTeX syntax or code provided by AI tools was verified by testing compilation in LaTeX documents, and ensuring the resulting output was correctly formatted and matched the needs of the document. No ideas or text were newly generated by AI tools and presented in the document.

9 Appendix — Reflection

Reflection Questions Answered:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Andreas:

1. When writing this deliverable, our team worked well to synchronize our decisions between sections. While I primarily focused on the FMEA table, I was able to discuss decisions my teammates had made on other vital sections (such as System Boundaries and Components) with them, clarify uncertainties, and use the output of our discussions to guide my work.
2. One pain point I experienced during this deliverable was finding suitable failure modes and hazards for our software, since it is intended to be a video game (it does not have many obvious physical safety risks). This was resolved by considering other ways failure of the system could result in harm, such as it being inaccessible to users with photosensitive epilepsy, or colourblind users. It was also found that other failures of the system could result in harm (in the expanded sense used in this project), by frustrating users and degrading the user experience overall.
3. Some risks had been considered before, such as failures of the system resulting in corrupted user save data, or unexpected behavior (that could frustrate the user), but specific risks concerning colourblind users and users with photosensitive epilepsy were newly found and formalized. These risks were found by considering the ways in which failures of the system could cause harm to specific user groups, if they were not accounted for in the design (as these risks could be missed without careful consideration). This led to finding a physical safety hazard (the potential for epileptic seizures in those with photosensitive epilepsy), as well as a potential risk of preventing colourblind users from playing the game as others can.
4. One risk can involve the risk of preventing certain users from being able to use a software product (as discussed above, in regards to accessibility). This risk is important to consider because it can result in real harm; users

with specific conditions can feel alienated when they aren't able to use or enjoy software products that others do. Another risk involves the risk of users becoming confused or frustrated when using the software product (as discussed in the FMEA analysis, when users' saved progress is lost, or the user interface is difficult to understand). This must be considered, as if most users become frustrated when using a software product, it will fail commercially/publicly in favour of other products that are designed with the user in mind.

Humna:

1. As a group, we communicated well, and the team had a good understanding of the overall project. My role for the Hazard Analysis was to complete Section 3, System Boundaries and Components. For this section, I broke RealmForge into its different components and considered how they interact with one another and the external environment. During the previous stages of the project, our group had already discussed the game's overall functionality and the features we wanted to include. As a result, I used our previous discussions to identify and describe the system components. Our group was also organized throughout the process. We implemented a review system where team members created pull requests that were reviewed before merging. This helped ensure everyone's work was checked and any necessary changes were addressed before finalizing the deliverable.

2. The main pain point I experienced was making sure the components aligned with the team's shared vision for RealmForge. When reviewing my section, I realized that some of the ideas I had included were potential features rather than confirmed components of the game. However, through discussions with the group, I was able to go back and make changes where needed. This reinforced the importance of reviewing our work as a group and having discussions to maintain consistency and have a shared understanding of the project.

3. Some risks we identified before working on the Hazard Analysis included the game potentially crashing and interrupting game sessions. Another risk we considered was issues with input devices, such as the game not recognizing or responding to a user's input method. One major risk we identified while working on this document was issues with player progression and how players move through the game. Another important risk we discussed further was progress not being saved correctly. Since progress is an important part of RealmForge, this could significantly affect the player's experience if they end up losing their progress between gaming sessions and having to start over.

4. Two important non-physical software risks are loss of user data and issues with the overall game functionality. Loss of data is a major concern because players could lose their saved progress and have to restart parts

of the game. Game functionality issues are also important, as crashes or a lack of input detection could disrupt sessions and negatively affect the gameplay experience. Both risks are important and should be considered to ensure RealmForge provides a functional and enjoyable experience for users.

Christopher:

1. When writing this deliverable, our team worked effectively to split work up based on prior knowledge as well as having members working on relevant sections to review eachothers work to ensure information remains consistent between the document. This allowed us to quickly work through sections while still ensuring the quality of this document and we could use this to help guide development in the future.

2. A pain point I experienced while writing this deliverable was trying to determine how safety and security requirements relate to a system which is virtual and not safety critical. This was resolved by considering how those with medical conditions may respond to certain stimulus, the security of virtual data as well as tester data, and a broader definition of the these requirements like how we can ensure the safety of our system by preventing failure modes.

3. During initial planning, the team identified many obvious risks that come with game development such as system crashing, data corruption, and UI confusion which are all things we have experienced ourselves when playing games. Risks that were thought of during the creation of this document were flashing lights and information barriers which would affect people with medical conditions and different experiences to us.

4. Two types of risks software products are:

    - Dependency Risks - Software often depends on software and hardware created and provided by third parties. If these dependencies stop receiving support or are detected to have a critical failure, it puts your own software at risk.
    - Data Integrity - Software rely on a lot of stored information which has the risk of being corrupted with no way for the software to stop some forms of corruption (there are ways to mitigate the damage caused by this corruption). Corrupted data can lead to completely wrong results and possibility of unrecoverable information or a very time-consuming recovery process.

Andy:

1. Good communication and a quick reaction towards finishing FMEA. The The FMEA needed other sections before being written. The group members all communicated so well in having their parts done that the chart was relatively smooth.

2. My main pain point was comming up with critical assumptions without ignoring possible hazards. I had to be delicate and not assume too much regarding the critical assumptions. If I were to make too much, then major assumptions wouldn't have bee looked at.

3. Simple failure hazards such as the system or software crashing was thought beforehand. Other risks such as the user's mental well being were thought about later. These risks came about when think about when the group thought about extenal factors that were essentially not the software.

4. As stated before, mental risks regarding the user's mind should be considered. The group didn't want to indoctrinate any controversal topic. Quality control risks are also necessary; they prevent the software from causing any harm to the user's operating device.

Owen:

1. When writing this deliverable, I think the team communicated well when working on different sections. The team asked for help when needed, and noticed when certain portions of divided work relied on other parts. The intro, scope and purpose were all intertwined with the other sections, with discussion taking place to ensure things fit together.

2. Some pain points that were experienced during this deliverable were time management, and finding adequate content to discuss in a relatively low risk, non safety critical system. Since the project is a video game it was harder to find physical risks, and this was resolved by including hazards/risks that are not immediately obvious, such as colourblind accessibility and interactions with photosensitive epilepsy. The pain point of time management was mainly because of the deliverable going through reading week, where schedules are uncertain and much different from a typical school week. This was resolved by an increase in Microsoft Teams activity, ensuring everyone was in the loop.

3. Some risks were thought of before this document, including loss of save data, game crashes, and other unintended behavior. These were all immediately identifiable risks that came about when initial discussions of developing a game were happening. Other, more specific risks came about later (during this document), such as users with photosensitive epilepsy having seizures due to flashing lights (this was discussed as a possibility in lecture). Specific risks such as these required extra thought about the design of the game, and not just the implementation.

4. 2 types of risk (other than physical) for software products include:

    - Barriers to Entry - Various design choices can cause barriers to entry for some users. For example, if similar colours are used and intended to be differentiated, this can cause colourblind users not to be able to interact with the game as intended.

- Unsatisfied Users - Many design choices affect users' enjoyment of the game. Increased frustration/confusion in figuring out how to interact with the game can cause users to be unsatisfied with their experience, which can lead to consequences like negative game reputation/reviews or game refunds for paid games.